

Pranav Ray

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PROFILE

Systems-focused software engineer building distributed systems and backend infrastructure in Go, from custom binary multiplexers and tunneling engines to Raft consensus, gossip protocols, and P2P networking.

PROJECTS

tunneru: Self-Hosted Tunneling Engine

Live

- Built a zero-dependency ngrok alternative in raw Go with a custom 9-byte binary multiplexer, handling hundreds of concurrent HTTP streams over a single persistent TCP connection with zero heap allocations in the fast path.
- Engineered a self-hosted reverse proxy with subdomain routing, token authentication, and a BubbleTea TUI displaying live request logs, stream metrics, and per-session uptime counters.
- Gained 65+ GitHub stars, 17K+ impressions on launch; featured on Trendshift as a trending Go repository within days of release.

Tech: Go, TCP/Multiplexing, BubbleTea, Cobra, Next.js, TypeScript, Tailwind CSS, WebSockets, Docker

tobira: Distributed Rate Limiter

GitHub

- Built a distributed rate limiter in Go coordinated via a custom gossip protocol over UDP, no Redis, no central coordinator, with nodes discovering and syncing state directly with each other.
- Implemented state-based CRDTs (grow-only counter maps) for deterministic conflict-free state merging, ensuring global rate limit counts converge correctly even after network partitions and node failures.
- Engineered four pluggable rate limiting algorithms (fixed window, sliding window, token bucket, leaky bucket) hot-swappable at runtime, with heartbeat-based failure detection and a Next.js + D3 topology dashboard.

Tech: Go, UDP, Gossip Protocol, CRDT, WebSockets, Next.js, TypeScript, D3.js, Docker

hotaru: Raft Consensus Implementation

GitHub

- Implemented the Raft consensus algorithm from scratch in Go, covering leader election via randomized timeouts, log replication across cluster nodes, and commit guarantees backed by majority acknowledgment.
- Built a replicated key-value store on top of the Raft layer as the state machine, exposing GET/SET/DELETE via REST API with automatic leader redirection for writes.
- Engineered a Next.js + D3 dashboard visualizing live leader elections, log entry propagation, commit index advancement, and node failure/recovery in real time via WebSocket stream.

Tech: Go, Raft, net/rpc, WebSockets, Next.js, TypeScript, Tailwind CSS, D3.js

tsuna: P2P Video Sync CLI

GitHub

- Implemented raw UDP hole punching and STUN-based NAT traversal in Go from scratch, enabling two peers behind separate home routers to connect directly with no media server involvement.
- Engineered a custom NTP-style clock reconciliation algorithm targeting sub-100ms playback sync across peers, using continuous round-trip time compensation to calculate and correct clock offset between machines.
- Built a lightweight WebSocket signaling server for initial peer handshake coordination, the only server in the system, going hands-off permanently after the hole-punch exchange completes.

Tech: Go, UDP/P2P, STUN, WebSockets, Cobra, BubbleTea, Next.js, TypeScript, Docker, Railway

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Python, C/C++, Swift

Frontend: React, Next.js, Tailwind CSS, shadcn/ui, Framer Motion, GSAP, Three.js, SwiftUI

Backend: Node.js, Bun, Express.js, Convex, Prisma, REST APIs, WebSockets

Databases: MongoDB, PostgreSQL, Redis, Firebase, Supabase

Distributed Systems: Gossip Protocols, CRDTs, Raft Consensus, UDP Networking, NAT Traversal

Tools: Git, Vercel, Docker, Railway, Figma, Postman, Cobra, BubbleTea, Xcode

EDUCATION

Manipal University Jaipur, Jaipur, India

B.Tech in Computer Science

CGPA: 8.0 / 10

2024 – 2028